

HEAMOFER 20 (Iron Sucrose Injection USP 20 mg/ml)

Description

HEAMOFER 20 (Iron Sucrose Injection USP 20 mg/ml), solution for injection or concentrate for solution for infusion which is dark brown, non-transparent, aqueous solution.
Each 5 mL of HEAMOFER 20 (Iron Sucrose Injection USP 20 mg/ml) ampoule contains 100 mg iron as iron sucrose (iron (III)-hydroxide sucrose complex) corresponding to 20mg iron per mL.

Pharmacological Properties

Pharmacodynamics

The polynuclear iron (III)-hydroxide cores are superficially surrounded by a large number of non-covalently bound sucrose molecules resulting in a complex whose molecular mass (Mw) is approximately 43 kDa. This is sufficiently large to prohibit renal elimination. The resulting complex is stable and does not release ionic iron under physiological conditions. The iron in the polynuclear cores is bound in a similar structure as in the case of physiologically occurring ferritin.

Mechanism of Action

HEAMOFER 20 is an aqueous complex of poly-nuclear iron (III)-hydroxide in sucrose. Following intravenous administration, Heamofer is dissociated into iron and sucrose and the iron is transported as a complex with transferrin to target cells including erythroid precursor cells. The iron in the precursor cells is incorporated into haemoglobin as the cells mature into red blood cells.

Pharmacokinetics

Following intravenous injection of a single dose of Iron Sucrose Injection containing 100 mg iron in healthy volunteers, maximum iron levels, averaging 538 mmol/l, were obtained 10 min after injection. The volume of distribution of the central compartment corresponded well to the volume of plasma (approximately 3 litres).

The iron injected was rapidly cleared from the plasma, the terminal half-life being approx.. 6 h. the volume of distribution at steady state was about 8 Litres, indicating a low iron distribution in the body fluid. Due to the lower stability of iron sucrose in comparison to transferrin, a competitive exchange of iron to transferrin was observed. This resulted in iron transport of approx.. 31 mg iron/24 h.

Renal elimination of iron, occurring in the first 4 h after injection, corresponds to less than 5% of the total body clearance. After 24 h the plasma levels of iron were reduced to the pre-dose iron level and about 75% of the dosage of sucrose was excreted.

Indication

HEAMOFER 20 is indicated for the treatment of iron deficiency anaemia in the following indications:

- Where there is a clinical need for a rapid iron supply.
- In patients who cannot tolerate oral iron therapy or who are non-compliant
- In active inflammatory bowel disease where oral iron preparations are ineffective.

HEAMOFER 20 should only be administered where the indication is confirmed by appropriate investigations (e.g. Hb, serum ferritin, serum iron).

Recommended Dose

Intravenous drip infusion

HEAMOFER 20 must be diluted only in sterile 0.9% m/V sodium chloride solution:
• 100 mg iron (5 ml HEAMOFER 20) in maximum 100 ml sterile 0.9% m/V sodium chloride solution
• 500 mg iron (25 ml HEAMOFER 20) in maximum 500 ml sterile 0.9% m/V sodium chloride solution
For stability reasons, dilutions to lower HEAMOFER 20 concentrations are not permissible.
As infusion, maximum tolerated single dose per day given not more than once per week:
• Patients above 70 kg: 500 mg iron (25 ml HEAMOFER 20) in at least 3½ hours
• Patients of 70 kg and below: 7 mg iron/kg body weight in at least 3½ hours
Dilution must take place immediately prior to infusion and the solution should be administered as follows:
• 100 mg iron (5 ml HEAMOFER 20) in at least 15 minutes
• 200 mg iron (10 ml HEAMOFER 20) in at least 30 minutes
• 300 mg iron (15 ml HEAMOFER 20) in at least 1½ hours
• 400 mg iron (20 ml HEAMOFER 20) in at least 2½ hours
Before administration of the therapeutic dose in a new patient the first 20 mg iron in adults and in children with a body weight greater than 14 kg and half the daily dose (1.5 mg iron/kg) in children with a body weight less than 14 kg should be infused over 15 minutes as a test dose. If no adverse reactions occur, the remaining portion of the infusion can be administered at recommended speed.

Intravenous injection

HEAMOFER 20 can be administered undiluted by slow intravenous injection as follows:

- 100 mg iron (5 ml HEAMOFER 20) in at least 5 minutes
- 200 mg iron (10 ml HEAMOFER 20) in at least 10 minutes

Before administration of the therapeutic dose in a new patient a test dose of 1 ml HEAMOFER 20 (20 mg iron) in adults and in children with a body weight greater than 14 kg and half the daily dose (1.5 mg iron/kg) in children with a body weight less than 14 kg should be injected over 1 to 2 minutes. If no adverse reactions occur within a waiting period of 15 minutes, the remaining portion of the injection can be administered at recommended speed.

Injection into dialyser

HEAMOFER 20 may be administered during a haemodialysis session directly into the venous limb of the dialyser under the same conditions as for intravenous injection.

Calculation of dosage

The total cumulative dose of HEAMOFER 20, equivalent to the total iron deficit (mg), is determined by the haemoglobin level and body weight. The dose of HEAMOFER 20 must be individually determined for each patient according to the total iron deficit calculated with the following formula:

Total iron Deficit [mg]
= body weight [kg] X (target Hb – actual Hb) [g/l] x 0.24* + depot iron [mg]

Below 35 kg body weight: target Hb = 130 g/l and depot iron = 15 mg/kg body weight

35 kg body weight and above: target Hb = 150 g/l and depot iron = 500 mg

* Factor 0.24 = 0.0034 x 0.07 x 1000:

Iron content of haemoglobin ≈ 0.34%
Blood volume ≈ 7% of body weight
Conversion from g/l to mg/l = Factor 1000

Total number of ampoules HEAMOFER 20 to be administered (In ml) = $\frac{\text{Total iron deficit [mg]}}{20 \text{ mg/ml}}$

Body Weight [kg]	Total number of ampoules HEAMOFER 20 to be administered (1 ampoule of HEAMOFER 20 corresponds to 5 ml)			
	Hb 60 g/l	Hb 75 g/l	Hb 90 g/l	Hb 105 g/l
5	1.5	1.5	1.5	1
10	3	3	2.5	2
15	5	4.5	3.5	3
20	6.5	5.5	5	4
25	8	7	6	5.5
30	9.5	8.5	7.5	6.5
35	12.5	11.5	10	9
40	13.5	12	11	9.5
45	15	13	11.5	10
50	16	14	12	10.5
55	17	15	13	11
60	18	16	13.5	11.5
65	19	16.5	14.5	12
70	20	17.5	15	12.5
75	21	18.5	16	13
80	22.5	19.5	16.5	13.5
85	23.5	20.5	17	14
90	24.5	21.5	18	14.5

To convert Hb (mM) to Hb (g/l), multiply the former by 16.1145.
If the total necessary dose exceeds the maximum allowed single dose, then the administration has to be split, please see section Administration.

Calculation of dosage for iron replacement secondary to blood loss and to support autologous blood donation

The required HEAMOFER 20 dose to compensate the iron deficit is calculated according the following formulas:
If the quantity of blood lost is known:
The administration of 200 mg i.v. iron (= 10 ml HEAMOFER 20) results in an increase in haemoglobin which is equivalent to 1 unit blood (= 400 ml with 150 g/l Hb content).
Iron to be replaced [mg] = number of blood units lost x 200
Amount of HEAMOFER 20 needed [ml] = number of blood units lost x 10

If the Hb level is reduced:
Use the following formula considering that the depot iron does not need to be restored.
Iron to be replaced [mg] = body weight [kg] x 0.24 x (target Hb – actual Hb) [g/l]
e. g.: body weight 60 kg, Hb deficit = 10 g/l
⇒ ≈150 ml iron to be replaced
⇒ 7.5 ml HEAMOFER 20 needed

Normal posology

Adults and the elderly:
5–10 ml HEAMOFER 20 (100–200 mg iron) one to three times a week depending on the haemoglobin level.
Children:
There is limited data on children under study conditions. If there is a clinical need, it is recommended not to exceed 0.15 ml HEAMOFER 20 (3 mg iron) per kg body weight one to three times per week depending on the haemoglobin level.

Maximum tolerated dose

As injection, maximum tolerated dose per day, given not more than three times per week:
• 200 mg iron (10 ml HEAMOFER 20) injected over at least 10 minutes.

As infusion, maximum tolerated single dose per day given not more than once per week:

- Patients above 70 kg: 500 mg iron (25 ml HEAMOFER 20) in at least 3½ hours
 - Patients of 70 kg and below: 7 mg iron/kg body weight in at least 3½ hours
- The maximum tolerated single dose is 7 mg iron per kg body weight given once per week, but not exceeding 500 mg iron. Administration time and dilution ratio see section administration. The infusion times given in section administration. Must be strictly adhered to, even if the patient does not receive the maximum tolerated single dose.

Route of Administration

HEAMOFER 20 must only be administered by the intravenous route. This may be by a slow intravenous injection or by an intravenous drip infusion.
HEAMOFER 20 must not be used for intramuscular injection.
Before administration of the first therapeutic dose, a test dose should be given. If any allergic reactions or intolerance occurs during administration, the therapy must be stopped immediately.

Contraindications

The use of HEAMOFER 20 is contraindicated in cases of:
• hypersensitivity to the active substance, to Iron Sucrose Injection or any of its excipients
• known serious hypersensitivity to other parenteral iron products
• anaemia not attributable to iron deficiency
• Iron overload or disturbances in utilisation of iron.

Warning and Precautions

Parenterally administered iron preparations can cause hypersensitivity reactions including serious and potentially fatal anaphylactic/anaphylactoid reactions. Hypersensitivity reactions have also been reported after previously uneventful doses of parenteral iron complexes.

The risk is enhanced for patients with known allergies including drug allergies, including patients with a history of severe asthma, eczema or other atopic allergy.
There is also an increased risk of hypersensitivity reactions to parenteral iron complexes in patients with immune or inflammatory conditions (e.g. systemic lupus erythematosus, rheumatoid arthritis).

HEAMOFER 20 should only be administered when staff trained to evaluate and manage anaphylactic reactions is immediately available, in an environment where full resuscitation facilities can be assured. Each patient should be observed for adverse effects for at least 30 minutes following each Iron Sucrose Injection. If hypersensitivity reactions or signs of intolerance occur during administration, the treatment must be stopped immediately. Facilities for cardio respiratory resuscitation and equipment for handling acute anaphylactic/anaphylactoid reactions should be available, including an injectable 1:1000 adrenaline solution. Additional treatment with antihistamines and/or corticosteroids should be given as appropriate.

In patients with liver dysfunction, parenteral iron should only be administered after careful risk/benefit assessment. Parenteral iron administration should be avoided in patients with hepatic dysfunction where iron overload is a precipitating factor, in particular Porphyria Cutanea Tarda (PCT). Careful monitoring of iron status is recommended to avoid iron overload.

Parenteral iron must be used with caution in case of acute or chronic infection. It is recommended that the administration of iron sucrose is stopped in patients with ongoing bacteraemia. In patients with chronic infection a risk/benefit evaluation has to be performed, taking into account the suppression of erythropoiesis.
Hypotensive episodes may occur if the injection is administered too rapidly. Allergic reactions, sometimes involving arthralgia, have been more commonly observed when the recommended dose is exceeded.
Paravenous leakage must be avoided because leakage of Iron Sucrose Injection at the injection site may lead to pain, inflammation, tissue necrosis and brown discoloration of the skin.

Effects on ability to drive and use machines

In the case of symptoms of dizziness, confusion or light headedness following the administration of HEAMOFER 20, patients should not drive or use machinery until the symptoms have ceased.

Incompatibilities

Iron Sucrose must only be mixed with sterile 0.9% m/V sodium chloride solution. No other solutions and therapeutic agent should be used as there is the potential for precipitation and/or interaction. The compatibility with containers other than glass, polyethylene and PVC is not known.

Interaction with Other Medicaments

As with all parenteral iron preparations, HEAMOFER 20 should not be administered concomitantly with oral iron preparations since the absorption of oral iron is reduced. Therefore an oral iron therapy should at least be started 5 days after the last injection of HEAMOFER 20.

Pregnancy and Lactation

There are no adequate and well-controlled trials of HEAMOFER 20 in pregnant women. A careful risk/benefit evaluation is therefore required before use during pregnancy and Iron Sucrose should not be used during pregnancy unless clearly necessary.

Iron deficiency anaemia occurring in the first trimester of pregnancy can in many cases be treated with oral iron. Treatment with Iron Sucrose Injection should be confined to second and third trimester if the benefit is judged to outweigh the potential risk for both the mother and the foetus.
Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development. Data on a limited number of exposed human pregnancies indicated no adverse effects of Iron Sucrose on pregnancy or on the health of the foetus/newborn child.

Non metabolized Iron Sucrose is unlikely to pass into the mother's milk. No well-controlled clinical studies are available to date. Animal studies do not indicate direct or indirect harmful effects to the nursing child.

Side Effects

The most frequently reported adverse drug reactions (ADRs) of Iron Sucrose were transient taste perversion, hypotension, fever and shivering, injection site reactions and nausea. Non-serious anaphylactoid reactions occurred rarely.
In general anaphylactoid reactions are potentially the most serious adverse reactions.

Nervous system disorders

Common: transient taste perversions (in particular metallic taste).
Uncommon: headache, dizziness.
Rare: paraesthesia, syncope, loss of consciousness, burning sensation.

Cardio-vascular disorders

Uncommon: hypotension and collapse, tachycardia and palpitations.
Rare: hypertension.

Respiratory, thoracic and mediastinal disorders

Uncommon: bronchospasm, dyspnoea.

Gastrointestinal disorders

Uncommon: nausea; vomiting, abdominal pain, diarrhoea.

Skin and subcutaneous tissue disorders

Uncommon: pruritus, urticaria, rash, exanthema, erythema.

Musculoskeletal, connective tissue and bone disorders

Uncommon: muscle cramps, myalgia.

General disorders and administration site disorders

Uncommon: fever, shivering, flushing, chest pain and tightness. Injection site disorders such as superficial phlebitis, burning, swelling.

Rare: arthralgia, peripheral oedema, fatigue, asthenia, malaise, feeling hot, oedema.

Immune system disorders

Rare: anaphylactoid reactions.
Moreover, in spontaneous reports the following adverse reactions have been reported:
Isolated cases: reduced level of consciousness, light-headed feeling, confusion, angio-oedema, swelling of joints, hyperhidrosis, back pain, bradycardia, chromaturia.

Symptoms and Treatment of Overdose

Over dosage can cause acute iron overdosing which may manifest itself as haemosiderosis. Over dosage should be treated, if required, with an iron chelating agent.

Storage Condition

Storage before opening: Store in original carton. Do not store above 30°C. Do not Freeze. Protect from Light.
Storage Condition after opening of the container: Not applicable, as it's a single ampoule 5ml. Not to store after opening.
Storage Condition after dilution with sterile 0.9% m/V sodium chloride solution: Store below 30°C.

Shelf Life

Shelf life of the product as packaged for sale: 2 years.
Shelf life after first opening of the container: From a microbiological point of view, the product should be used immediately.
Shelf life after dilution with sterile 0.9% m/V sodium chloride solution to a concentration of 1mg of elemental iron per mL: **Chemical and physical in use stability has been demonstrated for 12hours at 30°C.** From a microbiological point of view, the product should be used immediately after dilution with sterile 0.9% m/V sodium chloride solution. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

Presentation

5 ml USP Type I clear glass ampoule.

Manufactured by:

Gland Pharma Limited
Survey No. 143-148, 150 &151,
Near Gandimaisamma Cross Roads,
D.P. Pally, Dundigal Post,
Dundigal- Gandimaisamma Mandal ,
Medchal – Malkajiri District,
Hyderabad – 500043,
Telangana, India

Product Registration Holder:

UNIMED SDN BHD
53, Jalan Tembaga SD 5/2B,
Bandar Sri Damansara,
52200 Kuala Lumpur, Malaysia.

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PRODUCT NAME	Heamofer 20 (Iron sucrose Inj. USP)	ARTWORKS No.	LEA-019748-01
COUNTRY	Malaysia	SUPERSEDES No.	LEA-019748-00
CUSTOMER	Unimed SDN BHD	PANTONE SHADE No's	Black
DIMENSIONS	Open Dimension: 380 x 120 ± 2 mm (L x W) Folding Dimension: 48 x 60 ± 2 mm (L x W)	VARNISHING/LAMINATE	NA
SUBSTRATE	Maplitho paper (60 gsm ± 10%)	MODE OF SUPPLY	Bundles 25's/50's/100's